

Question Number	Answer	Additional Guidance	Mark
3(a)	A description that includes the following points: • find a suitable method for touching the external gills (1) • find suitable conditions that allow the sea slugs to be active (1) • find a suitable method to decide when the gills have re-emerged (1)	Ignore stated methods – credit is for finding out Accept eg. temperature, light intensity	Exp (3)

Question Number	Answer	Additional Guidance	Mark
3(b)	An answer that includes ten of the following points: • clear statement of the dependent variable e.g. (time for) re-emergence of gills (1) • <u>description of method</u> to provide different number of touches (1) • <u>description of method</u> for ensuring same intensity of each touch stimulation (1) • <u>description of method</u> for determining complete re-emergence of gills (1) • the method of touching the gills must not cause pain / damage (1) • • the animals are given time to acclimatise {before stimulation begins / between touches} (1) • variables that need to be taken into account (2) • description of how these variables are controlled (2) • repeat whole experiment with different animals (1)	Accept description of frequency of touches eg every 5 minutes eg glass rod of same diameter same person using same glass rod to touch with the same force eg measure / sketch gills at start and refer to this clear statement required eg water temperature, pH, salinity, depth of water, light intensity, substrate eg source / age / sex / species of Aplysia Accept repeat with the same animal (after suitable interval)	(10) exp

Question Number	Answer	Additional Guidance	Mark
3(c)	- table with headings and units (1) - means calculated from repeats (1) - line graph format with labelled axes (1) - use of an appropriate correlation statistical test (1)	Headings: time for re-emergence (accept minutes or seconds) and number of touches Accept description of table Accept statement or column in table Accept medians in place of mean Accept description of graph Accept named test eg Spearman Rank	(4) exp

Question Number	Answer	Additional Guidance	Mark
3(d)	- difficult to standardise the touch affecting the withdrawal reflex (1) - possible errors in determining full re-emergence (1) - difficult to control {age of animal / surface area of gills} (1) - difficult to control other possible stimuli which may cause reflex (1)	eg. pressure / force accept sex of Aplysia eg. sound, shadows, vibrations, water currents	(4) exp

(Total for question 3 = 21)